Vipul Sapra

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EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: Data Warehousing, Dimensional Data Modeling, Star/Snowflake Schemas, SCD Type 1/2, Batch and Incremental ELT/ETL, Data Governance, Query Optimization, CI/CD, Data Quality Audits, Predictive Modeling, Anomaly Detection

Programming Languages & Tools: Snowflake, AWS (S3, Redshift), Azure (ADF, Synapse, SQL), Python, PySpark, Apache Spark, SQL, Power BI, Tableau, Airflow, Git

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

Senior Data Engineer - Cloud Data Platforms & Analytics — **May 2022 – Present**

- **Architected high-performance Snowflake and AWS data models** Designed and optimized dimensional data modeling, star/snowflake schemas, and staging/curated data layers to support scalable analytics and reporting pipelines, matching Amex's rigorous data architecture standards.
- **Engineered robust batch and incremental ELT workflows** Developed scalable data pipelines using Python, SQL, and AWS cloud services (S3, Redshift) along with Azure Data Factory and Databricks, ensuring reliable ingestion and transformation of high-throughput operational datasets.
- **Implemented automated data quality validation frameworks** Deployed proactive anomaly detection systems and automated data quality audits with built-in firewalls, reducing operational turnaround by over 80% and ensuring 99.9% data availability SLAs.
- **Optimized Snowflake compute resources and query performance** Drove performance optimization across enterprise data warehousing layers by tuning complex SQL stored procedures, managing storage strategies, and improving query execution times.
- **Spearheaded enterprise data governance and CI/CD automation** Managed production runbooks, Airflow orchestration, and metadata documentation while leveraging agentic AI tools and CI/CD pipelines to accelerate secure, compliant data asset delivery by 40%.

Master of Science in Information Systems & Graduate Certificate in Data Science — **May 2022 – Present**

- **Completed rigorous advanced coursework in information systems** Achieved a 3.95/4.0 cumulative GPA at the University of Cincinnati, focusing on modern database architectures, enterprise data warehousing, and systems analysis.
- **Specialized in foundational data science and analytics** Built robust academic foundations in predictive modeling, statistical analysis, and large-scale data processing methodologies.
- **Applied theoretical concepts to real-world datasets** Engineered end-to-end data pipelines and machine learning prototypes as part of graduate research and technical projects.

Bachelor of Engineering in Petrochemical — **May 2022 – Present**

- **Developed strong analytical and quantitative problem-solving** Completed rigorous engineering coursework at Pune University MIT focusing on complex system dynamics, process optimization, and numerical analysis.
- **Mastered complex engineering mathematics and logic** Applied rigorous mathematical modeling and statistical methods to solve multifaceted optimization problems.
- **Cultivated cross-functional collaboration and technical execution** Collaborated on multi-disciplinary engineering projects requiring meticulous planning, data collection, and analytical evaluation.

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.
- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture — Pune, India

Test Engineering Analyst **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.
- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.
- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.